Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Menstrual Irregularities

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Under the Auspices of the IAP Action Plan 2022

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NORMAL MENSTRUAL CYCLES IN ADOLESCENCE

Introduction

Adolescence

Period of transition from childhood to adulthood and puberty happens during this period.

Menarche

- ☑ One of the markers of puberty in girls.
- ☑ Considered as an important event.
- ☑ Menarche occurs between the age of 10 and 15 years; however, can show variation.
- ☑ The age of menarche gets earlier as the nutritional, and economic conditions of the society improves.

Normal Cycles for Adolescence

- ☑ Duration every 20–45 days
- ☑ Flow ranging from 2 to 7 days
- ☑ Average flow of 35 mL blood loss or change of tampons/pads every 3–4 hours

Menstrual Disorders

- ☑ Common during adolescence.
- ☑ At least 75% of girls experience some problems with menstruation such as irregular, painful, and heavy menstrual bleeding, which are reasons for visiting the clinic frequently.
- ☑ Menstrual patterns are also influenced by a number of host and environmental factors.
- ☑ The medical history and examination will help giving clue to the underlying condition.

Menstrual Calendar (Fig. 1)

- ☑ Adolescents should be encouraged to chart their menstrual bleeding from the time of menarche.
- ☑ A menstrual calendar should be maintained, with dates and number of days of the cycle.
- ☑ This may improve the early identification of potential health concerns.
- ☑ This also helps clinicians to understand and differentiate normal and abnormal menstruation.
- ☑ By asking the menstruation history on all visits, clinicians emphasize the important role of menstrual patterns in reflecting overall health status.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	
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Type of flow: Bleeding—X Spotting—S

Fig. 1: Menstrual calender depicting the menstrual days and type of flow with the interval days between the cycle.

Common Menstrual Complaints

TABLE 1: Common menstrual complaints.

Term	*Definition*
Primary amenorrhea	No findings of secondary sexual characteristics by age 14 years or failure to menstruate within 3 years of the onset of physical changes of puberty
Secondary amenorrhea	Diagnosed after missing three periods after regular cycles had been established or 2 years after menarche
Dysmenorrhea	Painful menses not associated with pelvic pathology
Dysfunctional uterine bleeding	Prolonged excessive menstrual bleeding with no demonstrable organic cause; it is usually the result of anovulatory cycles
Polymenorrhea	Bleeding more frequently than every 18 days
Menorrhagia	Excessive uterine bleeding at regular intervals
Metrorrhagia	Intermenstrual bleeding

Clinical Management

History

- ☑ Family menstrual history from the mother because it is unlikely that the teen will have this information—good practice.
- ☑ Family history of fertility and bleeding problems, medications taken during pregnancy.
- ☑ *Taking teen history*:
  - Include serious illnesses and treatments
  - Age at menarche
  - Date of last normal menstrual cycle
  - Frequency, presence of menstrual pain, duration, and quantity of flow.
- ☑ History on chronic conditions that can affect menstruation are diabetes mellitus, inflammatory bowel disease, cystic fibrosis, and juvenile arthritis.
- ☑ History of medications intake, in particular that increase the frequency of bleeding.

Examination

Poor nutrition, both under and over, stress, and depression can also play disorder with regular cycles.

A pelvic examination:

- ☑ May not be necessary if the history and physical have determined the cause of the complaint.
- ☑ Should be done if there is any concern of a mass, alarm about the presence or normalcy of the vagina, cervix, or uterus.
- ☑ The pediatrician may wish to defer the pelvic examination to a colleague who is more experienced in doing pelvic examinations.

Dysmenorrhea

Primary dysmenorrhea commonly occurs as cycles become ovulatory. The basis of therapy is education and reassurance.

Secondary dysmenorrhea:

- ☑ Usually occurs later, well after the establishment of regular ovulatory cycles.
- ☑ Can result from such conditions as sexually transmitted diseases (STDs), pregnancy complications, endometriosis, or uterine or vaginal congenital anomalies.
- ☑ Treated by addressing the underlying etiology.

TABLE 2: Common treatment of dysmenorrhea.

Drug	*Dose*
Acetaminophen	650 mg qid
Ibuprofen	400 mg qid
Mefenamic acid (Ponstan, Parke Davis)	500 mg initially then 250 mg qid
Naproxen sodium	550 mg initially then 275 mg qid
Naproxen	500 mg initially then 250 mg qid

Amenorrhea

- ☑ Primary amenorrhea is uncommon.
- ☑ Pregnancy testing should be done in all teens who present with amenorrhea.
- ☑ The next step is to test thyroid-stimulating hormone (TSH), luteinizing hormone (LH), follicle-stimulating hormone (FSH), and dehydroepiandrosterone sulfate (DHEAS) levels.
- ☑ If the levels are normal, a challenge with medroxyprogesterone acetate 10 mg daily for 5 days is indicated.
- ☑ The teen should have withdrawal bleeding 2–7 days after this trial. Any bleeding is considered positive.
- ☑ With normal TSH and prolactin levels, this test confirms a normal hypothalamic-pituitary-ovarian axis and an intact reproductive tract.
- ☑ In those who fail to have withdrawal bleeding or who have abnormal laboratory tests, evaluation by an endocrinologist or gynecologist with an interest in pediatric and adolescent gynecology is indicated.

Dysfunctional Uterine Bleeding

- ☑ A complete history should include details about sexual activity, genital trauma, medications taken, and systemic illnesses.
- ☑ The most common cause of heavy menstrual bleeding is anovulation.
- ☑ If the bleeding time is abnormal, further evaluation for von Willebrand should be done.
- ☑ Depending on the history and physical examination, a thyroid profile as well as virilization tests (LH, FSH, prolactin, DHEAS, and free testosterone) may need to be assessed.
- ☑ The treatment of dysfunctional uterine bleeding is dependent on the severity of the bleeding.

Treatment

TABLE 3: Management of dysfunctional uterine bleeding.

Type	*Management*
Mild (normal hemoglobin)	☑ Menstrual calendar ☑ Reassurance ☑ Monitor iron status and follow every 1–2 months
Moderate (mild anemia; no hemodynamic instability)	☑ Oral contraceptives with norgestrel-ethinyl estradiol (Ovral, Wyeth-Ayerst) or norethindrone–mestranol (Norinyl, Searle) containing 50 µg of estrogenic component is used as follows [Gravol (Carter Horner) may be needed] – One pill qid × 4 days – One pill tid × 3 days – One pill bid × 7 days – One pill qid × 14–21 days then stop pills to allow withdrawal bleed. Then cycle with oral contraceptives for 3 months. ☑ Monitor iron status
Severe (hemoglobin below 90; hemodynamically unstable)	☑ Admit to hospital ☑ Fluids as needed ☑ Check for coagulopathy ☑ Transfusion as needed clinically and if bleeding cannot be stopped quickly ☑ *High dose estrogen:* – Orally—50 µg of norgestrel-ethinyl estradiol every 4 hours with an antiemetic until bleeding decreases then every 6 hours then every 8 hours – Intravenous—25 mg of estrogen (Premarin, Wyeth Ayerst) every 4 hours with an antiemetic until bleeding stops ☑ Iron supplementation then cycle with combination oral contraceptives for 3 months ☑ Follow-up

Summary

- ☑ Normal adolescent anovulatory cycles cause menstrual irregularity, particularly in the first postmenarcheal year.
- ☑ We should not ignore abnormal degrees of menstrual cyclicity in adolescents as "physiological" variations of normal.
- ☑ Evaluation is suggested when abnormal menstrual function persists ≥1 year or is associated with other symptoms or signs, like hirsutism, etc.

Further Reading

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Croup in Children

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